Consolidated Information on Surya Namaskar (SN) for reference and analysis

1. SN-base

We started with the figure below from [1] and tried to find the variants.

Fig. 1. Illustration of 12 linked-steps of Sun Salutation

Supporting Info: Standard as a base.

However, we chose the SN in Figure 2 which is followed at IIT-BHU from where we would be collecting sensing data. Hence, we are using it as our base SN version.

Fig. 2. Pictorial representation of Suryanamaskar **(Sivananda Yoga Vedanta Centre tradition):** Pictorial representation of the flow of postures of Surya namaskar as per Sivananda Yoga Vedanta Centre tradition. 1. Pranamasana; 2. Hasta utthanasana; 3. Pada hastasana; 4. Ashwasanchalanasana; 5. Phalakasana or Chaturanga Dandasan; 6. Ashtanga namaskara; 7. Bhujangasana; 8. Adho mukha svanasana or Parvatasan; 9. Ashwasanchalanasana; 10. Pada hastasana; 11. Hasta utthanasana; 12. Pranamasana.

Names of each pose and their corresponding hymns in the SN base.

1. Pranam-asana (prayer pose) Om Hraam/Om Mitraya Namah Anahata
2. Hasta Utthan-asana (raised arms pose) Om Hreem/Om Ravaye Namah Vishuddhi
3. **Padahast-asana** (hand to foot pose) Om Hroom/Om Suryaya Namah Swadhisthana
4. Ashwa Sanchalan-asana (equestrian pose) Om Hraim/Om Bhanave Namah Ajna
5. Chaturanga Dandasan (mountain pose) Om Hraum/Om Khagaya Namah Vishuddhi [also called Phalakasana]

6. Ashtanga Namaskara (salute with eight parts or points) Om Hrah/Om Pushne Namah Manipura
7. **Bhujang-asana** (cobra pose) Om Hraam/Om Hiranyagarbhaya Namah Swadhisthana
8. Parvat-asana (mountain pose) Om Hreem/Om Marichaye Namah Vishuddhi [also called Adho mukha svanasana]

9. Ashwa Sanchalan-asana (equestrian pose) Om Hroom/Om Adityaya Namah Ajna
10. **Padahasta-sana** (hand to foot pose) Om Hraim/Om Savitre Namah Swadhisthana
11. Hasta Utthan-asana (raised arms pose) Om Hraum/Om Arkaya Namah Vishuddhi
12. Pranam-asana (prayer pose) Om Hrah/Om Bhaskaraya Namah Anahata

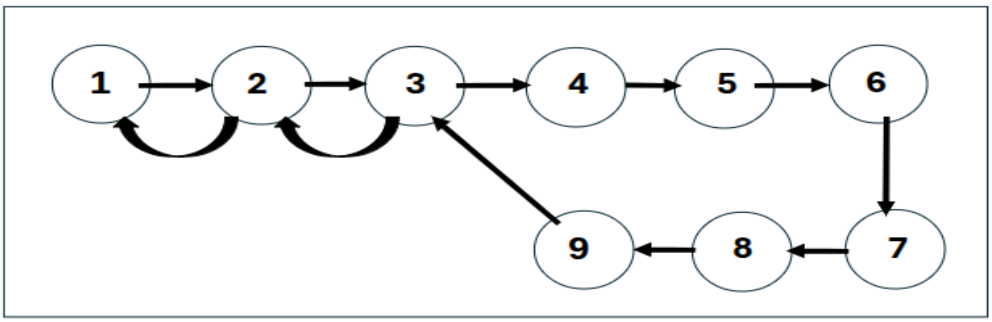

Fig. 3**. Schematic representation** of surya namaskar postures **(Sivananda Yoga Vedanta Centre tradition).**

Fig. 4. (Sivananda Yoga Vedanta Centre tradition) :- Sun Salutation.

2. SN variants

Variant-1: Krishnamacharya Vinyasa Suryanamaskar - 13-steps.

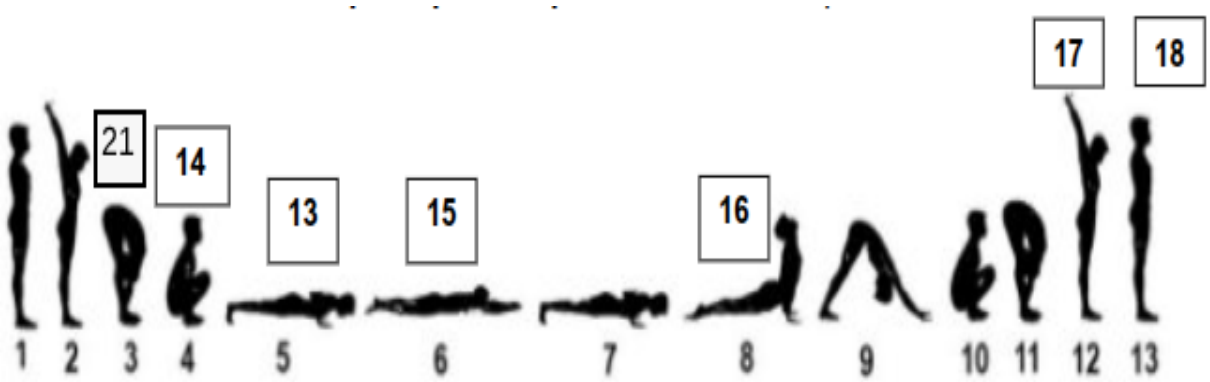

Fig. 5. Pictorial representation of Krishnamacharya Vinyasa Suryanamaskar. Pictorial representation of the flow of postures of Suryanamaskar as per Sri Krishnamacharya Vinyasa tradition. 1. Samasthiti/pranamasana 2. Tadasana/hasta uthanasana, 3. Uttanasana/pada hasthasana, 4. Utkatasana (hip stretch pose), 5. Chaturanga dandasana. 6. Danda samarpana, 7. Chaturanga dandasana, 8. Urdhwa mukha swanasana, 9. Adhomukha swanasana (Parvatasana), 10. Utkatasana, 11. Uttanasana (Pada hasthasana), 12. Tadasana, 13. Samasthiti. Picture adapted and modified from Ramaswami S. The complete book of Vinyasa yoga.

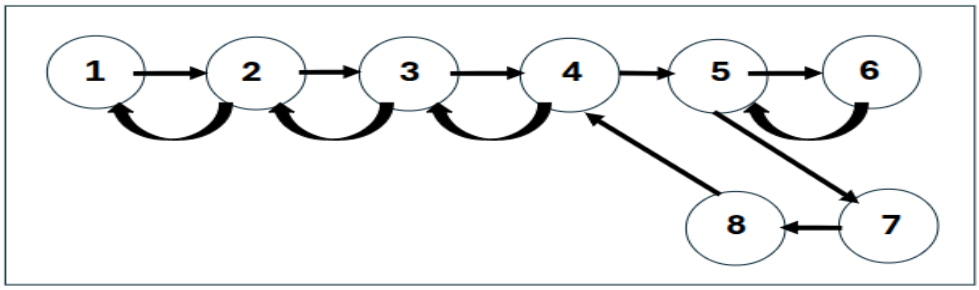

Fig. 6**. Schematic representation** of surya namaskar postures **(Krishnamacharya Vinyasa Suryanamaskar).**

Variant-2: Bihar School of Yoga tradition - 12-steps.

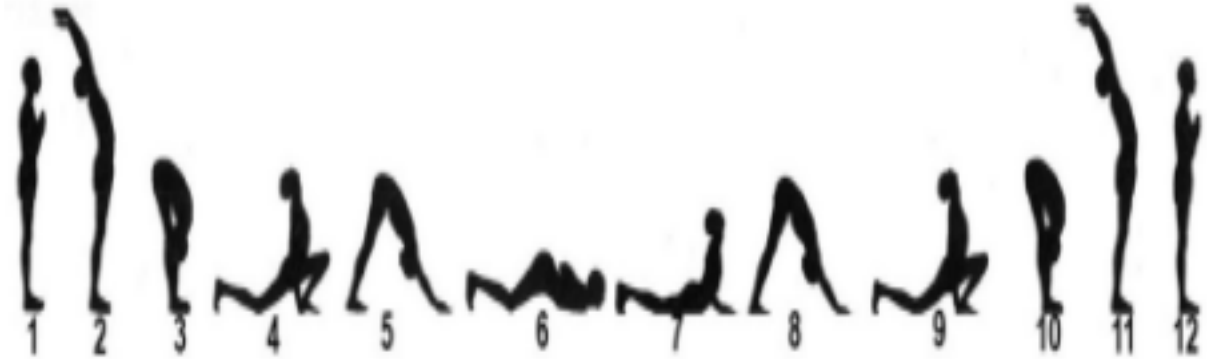

Fig. 7. Pictorial representation of Suryanamaskar postures (Bihar School of Yoga tradition): Final position of each step of Suryanamaskar of Bihar school of yoga tradition. Numbers (1e12) represents the number of each step. Picture obtained and modified from Swami Satyananda Saraswathi, Asana Pranayama Mudra Bandha.

Graphical representation of movement / workflow marking unique poses.

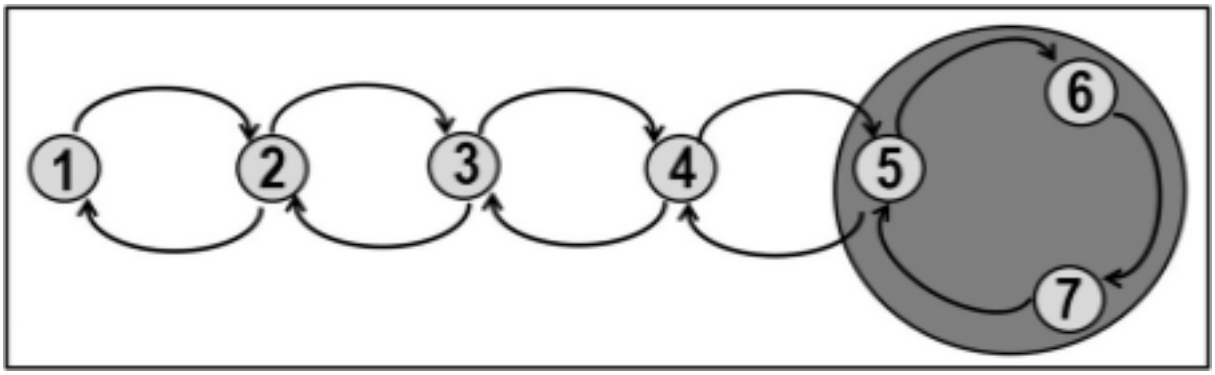

Fig. 8. **Schematic representation** of surya namaskar postures **(Bihar School of Yoga tradition):** Schematic of steps of Suryanamaskar of Bihar school of Yoga tradition numbered as per the sequence. Arrow indicates the flow of steps in the respective direction. 1. Pranamasana; 2. Hasta utthanasana; 3. Pada hastasana; 4. Ashwasanchalanasana; 5. Parvatasana; 6. Ashtanga namaskara; 7. Bhujangasana. Steps 5, 6 & 7 highlighted within the circle by themselves fulfill the steps of Dandaal practice.

Variant-3: Swami Vivekananda Kendra tradition. Uniqueness - poses 5 and 8 are different.

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Fig. 9. Pictorial representation of Suryanamaskar **(Swami Vivekananda Kendra tradition):** Pictorial representation of the flow of postures of Suryanamaskar as per Swami Vivekananda Kendra tradition. 1. Hastha uthanasana; 2. Padahathasana; 3. Ashwasanchalanasana; 4. Phalakasana or Dandasana; 5. Shashankasana; 6. Shashtanga namaskara; 7. Bhujangasana; 8. Parvatasana; 9. Shashankasana; 10. Ashwasanchalanasana; 11. Padahathasana; 12. Pranamasana.

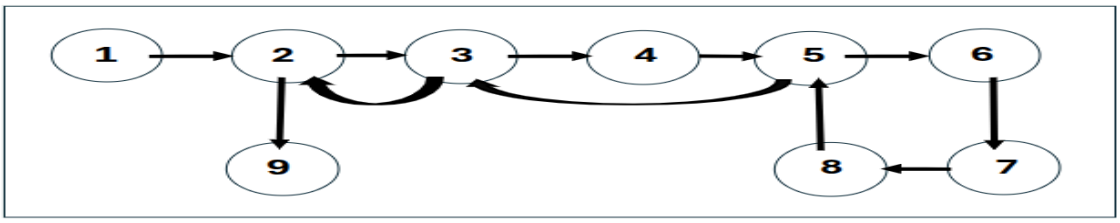

Fig. 10**. Schematic representation** of surya namaskar postures **(Swami Vivekananda Kendra tradition).**

3. Comments

Name of the postures in Suryanamaskar, respective Beeja Mantras/Mantras to be chanted, and corresponding Chakras to be concentrated. Steps Name of the Posture Beeja Mantra/Mantra Associated Chakra

1. Pranam-asana (prayer pose) Om Hraam/Om Mitraya Namah Anahata
2. Hasta Utthan-asana (raised arms pose) Om Hreem/Om Ravaye Namah Vishuddhi
3. **Padahast-asana** (hand to foot pose) Om Hroom/Om Suryaya Namah Swadhisthana
4. Ashwa Sanchalan-asana (equestrian pose) Om Hraim/Om Bhanave Namah Ajna
5. Parvat-asana (mountain pose) Om Hraum/Om Khagaya Namah Vishuddhi
6. Ashtanga Namaskara (salute with eight parts or points) Om Hrah/Om Pushne Namah Manipura
7. **Bhujang-asana** (cobra pose) Om Hraam/Om Hiranyagarbhaya Namah Swadhisthana
8. Parvat-asana (mountain pose) Om Hreem/Om Marichaye Namah Vishuddhi
9. Ashwa Sanchalan-asana (equestrian pose) Om Hroom/Om Adityaya Namah Ajna
10. **Padahasta-sana** (hand to foot pose) Om Hraim/Om Savitre Namah Swadhisthana
11. Hasta Utthan-asana (raised arms pose) Om Hraum/Om Arkaya Namah Vishuddhi
12. Pranam-asana (prayer pose) Om Hrah/Om Bhaskaraya Namah Anahata
13. Chaturanga dandasana / Phalakasana
14. Utkatasana
15. Danda samarpana
16. Urdhwa mukha swanasana
17. Tadasana

18. Samasthiti
19. Shashankasana
20. Shashtanga namaskara
21. Uttanasana
22. Phalakasana

See more details of each SN pose in the spreadsheet - SN-Pose-Details

4. Details of Each Pose

Total number of uniques poses in SN across all variants is: **20**

Mapping of unique poses to CNY asana is shown below.

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PĀDA-HASTĀSANA
(The Hands-to-Feet Posture)

Pāda means feet, *hasta* means hands. Therefore, *Pāda-Hastāsana* means keeping the palms down towards the feet. This is also referred as *Uttānāsana*.

Sthiti: *Samasthiti* (Alert posture)

Technique

- Stand straight with feet 2 inches apart.
- Inhale slowly and raise the arms up.
- Stretch up the body from the waist.
- Exhale and bend forward until both palms rest on the ground.
- Stretch the back, to make it as straight as possible.
- Maintain this final posture for 10–30 seconds and breathe normally.
- Now inhale, come up slowly to the upright position and stretch the arms straight above the head.
- Exhale, slowly return to the starting position in the reverse order.
- Relax in *Samasthiti*.

Benefits

- This *āsana* makes the spine flexible, improves digestion, and helps in overcoming menstrual problems.

Caution

- Avoid this practice in case of cardiac disorders, vertebral and disc disorders, abdominal inflammation, hernia and ulcers, glucoma, myopia, and vertigo.

NOTE
- Those who are suffering with stiff back should bend according to their capacity.

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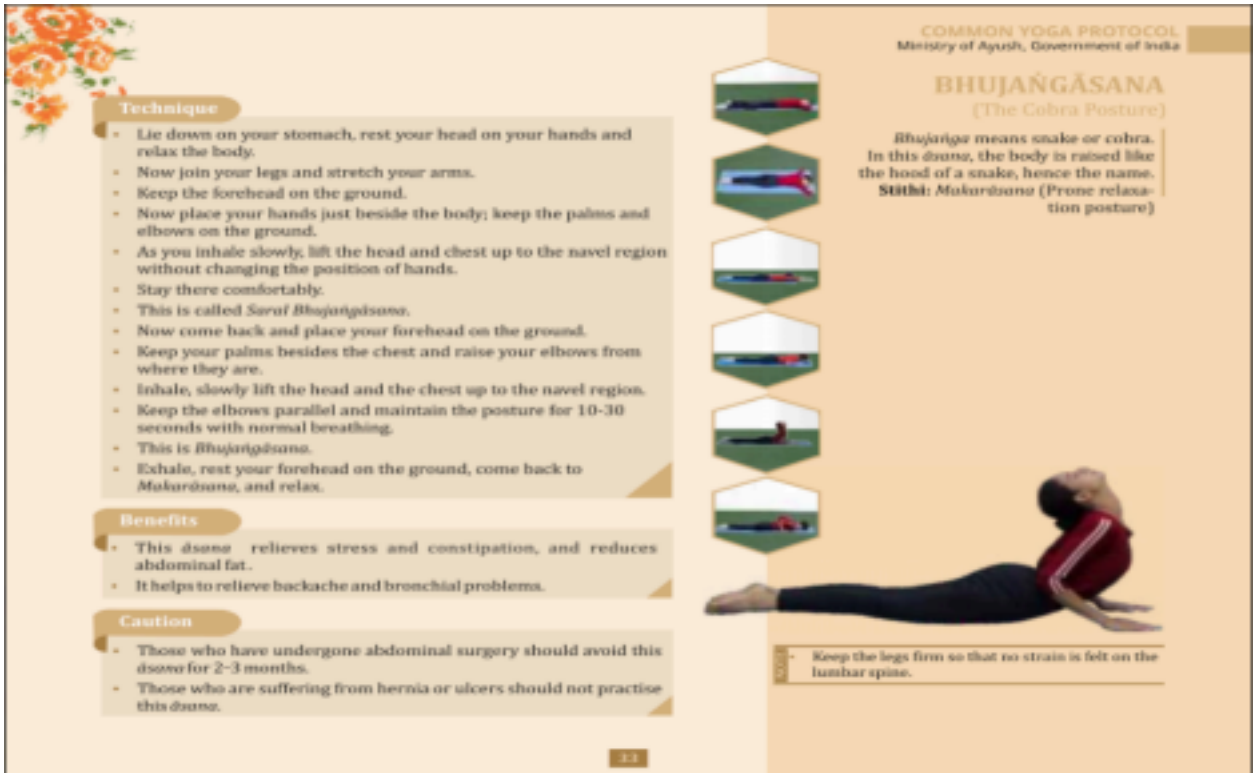
3 / 10. Padahast-asana (hand to foot pose), page no.- 22



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BHUJAŃGĀSANA
(The Cobra Posture)

Bhujaṅga means snake or cobra. In this *āsana*, the body is raised like the hood of a snake, hence the name.
Sthiti: *Makarāsana* (Prone relaxation posture)

Technique

- Lie down on your stomach, rest your head on your hands and relax the body.
- Now join your legs and stretch your arms.
- Keep the forehead on the ground.
- Now place your hands just beside the body; keep the palms and elbows on the ground.
- As you inhale slowly, lift the head and chest up to the navel region without changing the position of hands.
- Stay there comfortably.
- This is called *Sarpa Bhujaṅgāsana*.
- Now come back and place your forehead on the ground.
- Keep your palms besides the chest and raise your elbows from where they are.
- Inhale, slowly lift the head and the chest up to the navel region.
- Keep the elbows parallel and maintain the posture for 10-30 seconds with normal breathing.
- This is *Bhujaṅgāsana*.
- Exhale, rest your forehead on the ground, come back to *Makarāsana*, and relax.

Benefits

- This *āsana* relieves stress and constipation, and reduces abdominal fat.
- It helps to relieve backache and bronchial problems.

Caution

- Those who have undergone abdominal surgery should avoid this *āsana* for 2-3 months.
- Those who are suffering from hernia or ulcers should not practise this *āsana*.

NOTE
- Keep the legs firm so that no strain is felt on the lumbar spine.

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7. Bhujang-asana (cobra pose), page no. - 33

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A. STANDING POSTURES

3|YOGĀSANAS

TĀḌĀSANA

(Palm-tree Posture)

Tāḍa means palm tree or mountain.
This *āsana* helps one to attain stability and firmness and forms the base for all standing *āsans*.

Sthiti: *Samasthiti* (Alert posture)

Technique

- Stand with feet 2 inches apart.
- Inhale, lift your arms up to the shoulder level in the front.
- Interlock the fingers, and turn the wrist outwards. Now inhale, raise the arms up above your head.
- Raise the heels off the floor and balance on the toes as you raise your arms. Stay in this position for 10–30 seconds and breathe normally.
- Bring the heels down on exhale, release the interlock of the fingers and bring the arms down and come back to standing posture.

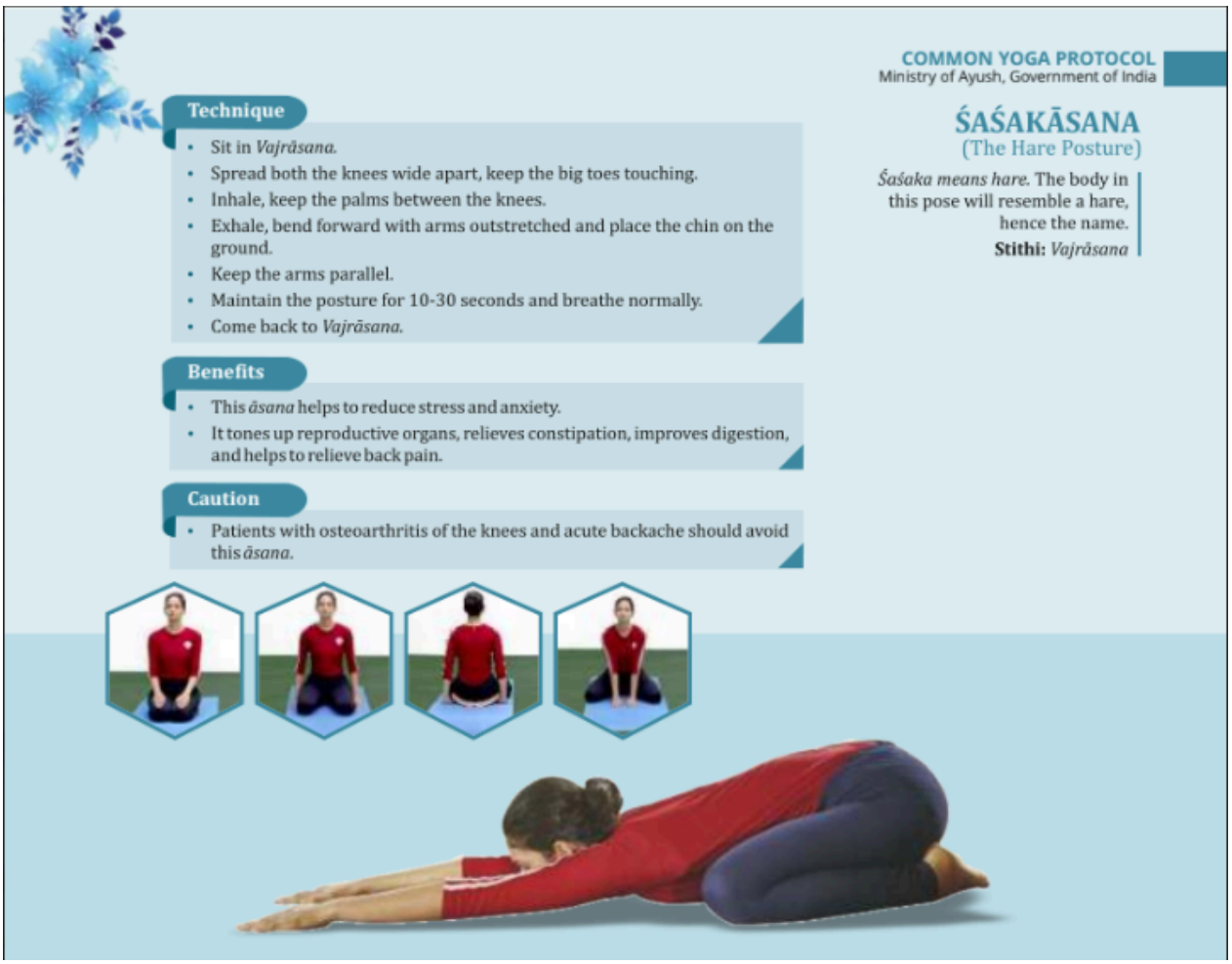
Benefits

- This *āsana* brings stability in the body, helps to clear up congestion of the spinal nerves, and corrects faulty posture.

Caution

- Avoid lifting the heels in case of arthritis, varicose veins, and vertigo.

17. Tadasana, page no. - 20



19. Shashankasana, page no. - 29

5. What More

SN has breath requirements (pranayam - exhale/ inhale) for each pose. We need to sense that as well because it is part of the full posture.

Sensor desiderata:

- Pose information
- Breath information
- Environmental information: water discharge

Other information that may be relevant: time of exercise, previous food and water intake

6. What More

SN has 22 poses across 4 variants. Only 4/22 = 18% were found in CYP.

Contributions:

a) SN consolidation - variants and poses
b) Identifying data issues in SN
c) Mapping unique poses to CYP

7. References and Credits

1. Fig. [1] Gupta, Ashish, and Hari Prabhat Gupta. "Yogahelp: Leveraging motion sensors for learning correct execution of yoga with feedback." IEEE Transactions on Artificial Intelligence 2.4 (2021): 362-371.
2. Fig. [2] - [6] Venkatesh, L. Prasanna, and S. Vandhana. "Insights on Surya namaskar from its origin to application towards health." Journal of Ayurveda and Integrative Medicine 13.2 (2022): 100530.
3. Section 4 from . Ayush document - CYP, https://yoga.ayush.gov.in/api/uploads/assets/cyp/Common%20Yoga%20Protocol%20Book-English.pdf?utm_source